

Cell-SELEX aptamer for highly specific radionuclide molecular imaging of glioblastoma in vivo.

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Glioblastoma (GBM) is the most frequent and aggressive primary adult brain tumor with poor prognosis. Epidermal growth factor receptor variant III (EGFRvIII) is the most common and highly oncogenic EGFR mutant in GBM. With the aim to generate specific molecular probes able to target EGFRvIII with high affinity, we selected four DNA aptamers (U2, U8, U19 and U31) specifically bound to U87-EGFRvIII cells that over expressed EGFRvIII with Kd values in the nanomole range by a cell-based systematic evolution of ligands by exponential enrichment (cell-SELEX) process. U87MG cells were introduced as control cells for counter selection. We further affirmed U2 and U8 identified EGFRvIII on the surface of target cells specifically. Then we radiolabeled U2 with ¹⁸⁸Re to serve as a molecular imaging probe and observed ¹⁸⁸Re -labeled U2 significantly targeted EGFRvIII over-expressing glioblastoma xenografts in mice. In conclusion, aptamers obtained from whole cell-SELEX strategy have great potential as molecular imaging probes that are probably beneficial to GBM diagnoses.

EGFR wild type antagonizes EGFRvIII-mediated activation of Met in glioblastoma.

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Epidermal growth factor receptor (EGFR)vIII is the most common EGFR mutant found in glioblastoma (GBM). EGFRvIII does not bind ligand, is highly oncogenic and is usually coexpressed with EGFR wild type (EGFRwt). EGFRvIII activates Met, and Met contributes to EGFRvIII-mediated oncogenicity and resistance to treatment. Here, we report that addition of EGF results in a rapid loss of EGFRvIII-driven Met phosphorylation in glioma cells. Met is associated with EGFRvIII in a physical complex. Addition of EGF results in a dissociation of the EGFRvIII-Met complex with a concomitant loss of Met phosphorylation. Consistent with the abrogation of Met activation, addition of EGF results in the inhibition of EGFRvIII-mediated resistance to chemotherapy. Thus, our study suggests that ligand in the milieu of EGFRvIII-expressing GBM cells is likely to influence the EGFRvIII-Met interaction and resistance to treatment, and highlights a novel antagonistic interaction between EGFRwt and EGFRvIII in glioma cells.

F25P preproinsulin abrogates the secretion of pro-growth factors from EGFRvIII cells and suppresses tumor growth in an EGFRvIII/wt heterogenic model.

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Extensive heterogeneity is a defining hallmark of glioblastoma multiforme (GBM) at the cellular and molecular levels. EGFRvIII, the most common EGFR mutant, is expressed in 24-67% of cases and strongly indicates a poor survival prognosis. By co-expressing EGFRvIII and EGFRwt, we established an EGFRvIII/wt heterogenic model. Using this approach, we confirmed that a mixture of EGFRvIII and EGFRwt at a certain ratio could clearly enhance tumor growth in vitro and in vivo compared with EGFRwt cells, thereby indicating that EGFRvIII cells promote tumor growth. Furthermore, we demonstrated that the EGFRvIII cells could support the growth of EGFRwt cells by secreting growth factors, thus acting as the principal source for maintaining tumor survival. F25P preproinsulin effectively reduced the concentrations of EGF, VEGF, and MMP-9 in the blood of tumor-bearing mice by competitively inhibiting the endoplasmic

reticulum signal peptidase and increased the overall survival in orthotopic models. Taken together, our results provided an effective therapy of F25P preproinsulin in the EGFRvIII/wt heterogenic model.

N6-Isopentenyladenosine Impairs Mitochondrial Metabolism through Inhibition of EGFR Translocation on Mitochondria in Glioblastoma Cells.

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Glioblastoma multiforme (GBM) is the most aggressive malignant brain tumor and is poorly susceptible to cytotoxic therapies. Amplification of the epidermal growth factor receptor (EGFR) and deletion of exons 2 to 7, which generates EGFR variant III (vIII), are the most common molecular alterations of GBMs that contribute to the aggressiveness of the disease. Recently, it has been shown that EGFR/EGFRvIII-targeted inhibitors enhance mitochondrial translocation by causing mitochondrial accumulation of these receptors, promoting the tumor drug resistance; moreover, they negatively modulate intrinsic mitochondria-mediated apoptosis by sequestering PUMA, leading to impaired apoptotic response in GBM cells. N6-isopentenyladenosine (i6A or iPA), a cytokinin consisting of an adenosine linked to an isopentenyl group deriving from the mevalonate pathway, has antiproliferative effects on numerous tumor cells, including GBM cells, by inducing cell death in vitro and in vivo. Here, we observed that iPA inhibits the mitochondrial respiration in GBM cells by preventing the translocation of EGFR/EGFRvIII to the mitochondria and allowing PUMA to interact with them by promoting changes in mitochondrial activity, thus playing a critical role in cell death. Our findings clearly demonstrate that iPA interferes with mitochondrial bioenergetic capacity, providing a rationale for an effective strategy for treating GBM.

Effects of combined radiosurgery and temozolomide therapy on epidermal growth factor receptor and variant III in glioblastoma multiforme.

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Glioblastoma multiforme (GBM) is a highly malignant and notably aggressive primary tumour. Variant III of the epidermal growth factor receptor (EGFRvIII) is one of the most common types of variants in GBM, and serves an important role in tumour invasion, proliferation and treatment resistance. In the present study, statistical analyses were performed on data from 57 patients with GBM, and polymerase chain reaction detection was conducted on the tumour tissues from 32 of these patients. The results indicated that the EGFRvIII mutation was significantly associated with tumour malignancy. Human GBMU87-EGFRvIII cell lines were cultured and treated with radiosurgery and temozolomide individually, or with combined radiosurgery and temozolomide treatment. In vitro and in vivo experimental methods were used to detect the expression levels of Ki-67 and EGFRvIII. As verified in the present study, the EGFRvIII mutation is positively correlated with the malignancy of tumours, and combined radiosurgery and temozolomide therapy may inhibit the invasion and proliferation abilities of U87-EGFRvIII more effectively than treatment alone.

Epidermal growth factor receptor variant III-induced glioma invasion is mediated through myristoylated alanine-rich protein kinase C substrate overexpression.

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Glioblastoma multiforme (GBM) is the most common and most malignant adult brain tumor. A characteristic of GBM is their highly invasive nature, making complete surgical resection impossible. The most common gain-of-function alteration in GBM is amplification, overexpression, and mutations of the epidermal growth factor receptor (EGFR). The constitutively activated mutant EGFR variant III (EGFRvIII), found in approximately 20% of GBM, confers proliferative and invasive advantage. The signaling cascades downstream of aberrant EGFR activation contributing to the invasive phenotype are not completely understood. Here, we show myristoylated alanine-rich protein kinase C substrate (MARCKS), previously implicated in cell adhesion and motility, contributes to EGFR-mediated invasion of human GBM cells. EGFRvIII-expressing or EGF-stimulated human GBM cells increased expression, phosphorylation, and cytosolic translocation of MARCKS in a protein kinase C- α -dependent manner. Down-regulation of MARCKS expression with small interfering RNA in GBM cells expressing EGFRvIII led to decreased cell adhesion, spreading, and invasion. Elucidation of mechanisms that promote EGFRvIII-mediated tumorigenesis in GBM, such as MARCKS, provides additional understanding and potential biological targets against this currently terminal human cancer.

Current Therapeutic Advances Targeting EGFR and EGFRvIII in Glioblastoma.

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Epidermal growth factor receptor (EGFR) and EGFRvIII analysis is of current interest in glioblastoma - the most common malignant primary CNS tumor, because of new EGFRvIII vaccine trials underway. EGFR activation in glioblastoma promotes cellular proliferation via activation of MAPK and PI3K-Akt pathways, and EGFRvIII is the most common variant, leading to constitutively active EGFR. This review explains EGFR and EGFRvIII signaling in GBM; describes targeted therapy approaches to date including tyrosine kinase inhibitor, antibody-based therapies, vaccines and pre-clinical RNA-based therapies, and discusses the difficulties encountered with these approaches including pathway redundancy and intratumoral heterogeneity.

[Targeting EGFRvIII for treatment of glioblastoma: From molecular mechanisms to clinical strategies].

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Glioblastoma is one of the most common intracranial malignant tumor and its initiation and progression are closely associated with epidermal growth factor receptor (EGFR). EGFR variant III (EGFRvIII) is a mutant EGFR and highly expressed in glioblastoma. EGFRvIII promotes the proliferation and invasiveness of glioblastoma cells and induces drug resistance by signaling networks.

Mapping of genomic EGFRvIII deletions in glioblastoma: insight into rearrangement mechanisms and biomarker development.

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Epidermal growth factor receptor (EGFR) variant III (vIII) is the most common oncogenic rearrangement in glioblastoma (GBM), generated by deletion of exons 2 to 7 of EGFR. The proximal breakpoints occur in variable positions within the 123-kb intron 1, presenting significant challenges in terms of polymerase chain reaction (PCR)-based mapping. Molecular mechanisms underlying these deletions remain unclear. We determined the presence of EGFRvIII and its breakpoints for 29 GBM samples using quantitative PCR, arrayed PCR mapping, Sanger sequencing, and whole genome sequencing (WGS). Patient-specific breakpoint PCR was performed on tumors, plasma, and cerebrospinal fluid (CSF) samples. The breakpoint sequences and single nucleotide polymorphisms (SNPs) were analyzed to elucidate the underlying biogenic mechanism. PCR mapping and WGS independently unveiled 8 EGFRvIII breakpoints in 6 tumors. Patient-specific primers yielded EGFRvIII PCR amplicons in matched tumors and in cell-free DNA (cfDNA) from a CSF sample, but not in cfDNA or extracellular-vesicle DNA from plasma. The breakpoint analysis revealed nucleotide insertions in 4 samples, an insertion of a region outside of the EGFR locus in 1, microhomologies in 3, as well as a duplication or an inversion accompanied by microhomologies in 2, suggestive of distinct DNA repair mechanisms. In the GBM samples that harbored distinct breakpoints, the SNP compositions of EGFRvIII and amplified non-vIII EGFR were identical, suggesting that these rearrangements arose from amplified non-vIII EGFR. Our approach efficiently "fingerprints" each sample's EGFRvIII breakpoints. Breakpoint sequence analyses suggest that independent breakpoints arose from precursor amplified non-vIII EGFR through different DNA repair mechanisms.

Epidermal growth factor receptor and PTEN modulate tissue factor expression in glioblastoma through JunD/activator protein-1 transcriptional activity.

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Hypoxia and necrosis are fundamental features of glioblastoma (GBM) and their emergence is critical for the rapid biological progression of this fatal tumor; yet, underlying mechanisms are poorly understood. We have suggested that vaso-occlusion following intravascular thrombosis could initiate or propagate hypoxia and necrosis in GBM. Tissue factor (TF), the main cellular initiator of coagulation, is overexpressed in GBMs and likely favors a thrombotic microenvironment. Epidermal growth factor receptor (EGFR) amplification and PTEN loss are two common genetic alterations seen in GBM but not in lower-grade astrocytomas that could be responsible for TF up-regulation. The most frequent EGFR mutation in GBM involves deletion of exons 2 to 7, resulting in the expression of a constitutively active receptor, EGFRvIII. Here, we show that overexpression of EGFR or EGFRvIII in human glioma cells causes increased basal TF expression and that stimulation of EGFR by its ligand, EGF, leads to a marked dose-dependent up-regulation of TF. In all cases, increased TF expression led to accelerated plasma coagulation in vitro. EGFR-mediated TF expression depended most strongly on activator protein-1 (AP-1) transcriptional activity and was associated with c-Jun NH(2)-terminal kinase (JNK) and JunD activation. Restoration of PTEN expression in PTEN-deficient GBM cells diminished EGFR-induced TF expression by inhibiting JunD/AP-1 transcriptional activity. PTEN mediated this effect by antagonizing phosphatidylinositol 3-kinase activity, which in turn attenuated both Akt and JNK activities. These mechanisms are likely at work in vivo, as EGFR expression was highly correlated with TF expression in human high-grade astrocytoma specimens.